

Ordering Mixed Rational and Irrational Quantities

Converting each quantity to decimal form, comparing a pair, and ordering a mixed list

Directions

- Fractions, roots and decimals cannot be compared as they are written. Put every quantity into decimal form first — two places is enough for everything here.
 - An irrational value never converts exactly, so use $\approx$ for it and $=$ for the exact ones.
 - Show the decimal you used for each quantity, not just the final answer.
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1. Write $\frac{5}{8}$ as a decimal.

Answer: _____

2. Give $\sqrt{11}$ to the nearest hundredth.

Answer: _____

3. Write $<$, $>$, or $=$ between the two quantities.

$$\frac{7}{9} \quad \text{_____} \quad 0.78$$

4. Write $\frac{17}{4}$ as a decimal.

Answer: _____

5. Write $<$, $>$, or $=$ between the two quantities.

$$\sqrt{17} \quad \underline{\hspace{1cm}} \quad 4.2$$

6. Put these three quantities in order from least to greatest.

$$\begin{array}{ccccc} \sqrt{5} & 2.3 & \frac{9}{4} & & \\ \underline{\hspace{1cm}} & < & \underline{\hspace{1cm}} & < & \underline{\hspace{1cm}} \end{array}$$

7. Give π to the nearest hundredth.

Answer: _____

- 8.** Write $<$, $>$, or $=$ between the two quantities. Two decimal places are not enough here — carry a third.

$$\pi \quad \underline{\hspace{1cm}} \quad \sqrt{10}$$

- 9.** Put these three quantities in order from least to greatest.

$$\begin{array}{ccc} \sqrt{2} & \frac{3}{2} & 1.4 \\ \underline{\hspace{1cm}} < \underline{\hspace{1cm}} < \underline{\hspace{1cm}} \end{array}$$

- 10.** Write $<$, $>$, or $=$ between the two quantities. Remember which way round the number line runs for negatives.

$$-\sqrt{9} \quad \underline{\hspace{1cm}} \quad -3.1$$

11. Put these four quantities in order from least to greatest.

$$\begin{array}{ccccccc} \sqrt{30} & & 5.6 & & \frac{27}{5} & & \sqrt{36} \\ \hline \rule{1.5cm}{0.4pt} & < & \rule{1.5cm}{0.4pt} & < & \rule{1.5cm}{0.4pt} & < & \rule{1.5cm}{0.4pt} \end{array}$$

12. **Challenge.** Nadia is asked to order $\sqrt{16}$, 3.9 and $\frac{15}{4}$ from least to greatest. She looks at the numbers as they are printed and answers “3.9, then $\frac{15}{4}$, then $\sqrt{16}$ ”.

- (a) Give the correct order, least to greatest.

$$\rule{1.5cm}{0.4pt} < \rule{1.5cm}{0.4pt} < \rule{1.5cm}{0.4pt}$$

- (b) Explain why the numbers as printed cannot be compared directly, and what has to happen first.